

KANDULA JITHENDRA SUBRAMANYAM

M.Tech AI & Data Science | Agentic AI | Multi-Agent Systems | Quantitative Finance | Full-Stack Dev

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

PROFESSIONAL SUMMARY

M.Tech Artificial Intelligence and Data Science candidate with hands-on experience in agentic AI, multi-agent systems, portfolio governance, quantitative finance, explainable AI, and full-stack development. Built an Agentic AI Portfolio Governance Chatbot and Multi-Agent Adaptive Portfolio Governance System using specialized agents for planning, data retrieval, technical analysis, market-regime detection, instability analysis, portfolio optimization, governance validation, explainability, and verification. Also engineered a production real-time voice-AI agent — sub-500ms latency, LiveKit/WebRTC, LangGraph multi-agent orchestration, and RAG-grounded responses — deployed live to power an interactive, voice-driven portfolio. Research manuscripts submitted to Elsevier EAAI and Springer conference proceedings are currently under review. Skilled in Python, LangChain/LangGraph, CVXPY, NetworkX, MongoDB, React, Next.js, Flask, REST APIs, LiveKit/WebRTC, financial risk analytics, and compliance-aware AI systems.

TECHNICAL SKILLS

AI, ML & Research	Artificial Intelligence, Machine Learning, Deep Learning, Agentic AI, Multi-Agent Systems, Explainable AI, Retrieval-Augmented Generation, Prompt Engineering
Finance & Optimization	Quantitative Finance, Portfolio Optimization, CVaR, Drawdown, Volatility, RSI, MACD, Bollinger Bands, Market Regime Detection, Compliance-Aware AI
Agentic AI & LLM Tools	LangChain, LangGraph, Groq, Ollama, Mistral-7B, Agent Memory, Verification Frameworks, Audit Logging, Source Grounding
Voice AI & Real-Time Systems	LiveKit, WebRTC, Deepgram (STT), ElevenLabs (TTS), Real-Time Data Channels, Sub-500ms Conversational Pipelines
Optimization & Graph Tools	CVXPY, CLARABEL, NetworkX, Risk-Aware Portfolio Construction, Graph-Based Reasoning
Programming & Data	Python, NumPy, Pandas, SciPy, Scikit-learn, SQL, MongoDB, Supabase, Git, GitHub
Full-Stack Development	React, Next.js, TypeScript, JavaScript, Flask, REST APIs, HTML, CSS, Gradio, AQICN API

PROFESSIONAL EXPERIENCE

Thesis Researcher – Agentic AI Portfolio Governance

Oct 2025 – Apr 2026

K J Somaiya College of Engineering | Mumbai

- Developed an Agentic AI Portfolio Governance Chatbot for stock portfolio management using a modular multi-agent architecture.
- Designed specialized agents for task planning, data retrieval, technical analysis, market-regime detection, instability analysis, portfolio optimization, governance validation, explainability, self-verification, and final response generation.
- Implemented audit logging, confidence scoring, evidence tracking, source grounding, anti-hallucination controls, and compliance-aware decision support to improve reliability of financial AI outputs.
- Built technical-analysis modules for RSI, MACD, Bollinger Bands, support/resistance, trend analysis, volatility, drawdown, and Bloomberg-style reporting.
- Integrated Python, LangChain/LangGraph, CVXPY, CLARABEL, NetworkX, MongoDB, Gradio, Ollama, Mistral-7B, YFinance, NumPy, Pandas, SciPy, and financial datasets.

Full-Stack Developer Intern

May 2025 – Aug 2025

ScholarRankAI | Remote

- Designed and delivered responsive React-based user interfaces for production workflows.
- Integrated third-party REST APIs and supported end-to-end frontend functionality.
- Participated in Agile stand-ups, sprint planning, debugging cycles, code reviews, and performance optimization.
- Reduced page-load bottlenecks through targeted frontend improvements and enhanced usability across application screens.

UI/UX Developer Intern

Mar 2024 – May 2024

MNJ Software Pvt. Ltd. | India

- Translated UI/UX wireframes into production-ready HTML, CSS, and JavaScript interfaces.
- Maintained visual consistency across screens and collaborated with backend engineers on frontend-backend integration.
- Improved page-load speed by 25% and increased user engagement metrics by 15% through targeted UI and performance enhancements.

PROJECTS

AI-Powered Agentic Voice Portfolio

2026 – Present

[Live Demo](#) | [GitHub](#)

- Engineered a production, real-time voice-AI portfolio achieving sub-500ms conversational latency, letting visitors talk through projects, research, and experience instead of browsing a static site.
- Architected a LangGraph multi-agent swarm — Greeter, Research, Engineering, and Booking specialists — orchestrating Deepgram (STT), Groq-hosted Llama-3.3-70B inference, and ElevenLabs (TTS) over LiveKit WebRTC audio and data channels.
- Built a RAG pipeline grounding 378 vector chunks from 6 research papers and technical docs to curb hallucination, so the agent answers research and engineering questions from verified sources.

- Enabled autonomous UI control: the agent drives DOM navigation, theme switching, and resume/document downloads in sync with its speech via WebRTC data-channel action dispatch. - Shipped as a two-tier production system — Next.js 15 / React 19 frontend on Vercel, Python LiveKit worker backend on Render — with Supabase logging session and per-turn latency metrics.	
Agentic AI Portfolio Governance Chatbot	Oct 2025 – Apr 2026
- Built a personal chatbot for stock portfolio management using a multi-agent AI framework with planning, data retrieval, technical analysis, regime detection, instability analysis, portfolio optimization, governance validation, explainability, verification, and response generation agents. - Implemented audit logs, confidence scores, evidence tracking, source grounding, anti-hallucination controls, and compliance-aware outputs. - Stack: Python, LangChain/LangGraph, CVXPY, CLARABEL, NetworkX, MongoDB, Gradio, Ollama, Mistral-7B, YFinance, NumPy, Pandas, SciPy.	
Multi-Agent Adaptive Portfolio Governance System	Jan 2026 – Apr 2026
- Designed a financial decision-intelligence system combining agentic orchestration, risk-aware portfolio construction, regime-aware adaptation, instability detection, optimization, explainability, governance checks, and response verification. - Built the system to support safer financial reasoning through evidence-backed responses, governance approval checks, transparent explanations, and audit-ready outputs; formed the basis for two manuscripts currently under review at Elsevier and Springer.	
Personalised AQI – Global Air Quality Forecasting System	Feb 2023 – Mar 2025
- Built a full-stack AQI forecasting platform using live AQICN.org data to predict city-level air quality and pollutant trends worldwide. - Implemented ML models for AQI, PM2.5, O3, and NO2 forecasting with explainable health-risk insights. - Designed a Next.js dashboard for real-time and predicted AQI trend visualization across cities, states, and countries. - Stack: Next.js, Python, Flask, MongoDB, Machine Learning, Explainable AI, AQICN API.	
Swarm Robots for Agriculture	Aug 2022 – Jul 2023
- Built a swarm robotics agriculture project using embedded systems, IoT sensors, autonomous coordination, and image-processing concepts. - Worked on plant disease detection, sensor-based monitoring, robot coordination, and agriculture-focused automation. - Combined electronics, communication systems, computer vision concepts, and embedded implementation for agricultural assistance.	

RESEARCH PAPERS / PUBLICATIONS

Multi-Agent Adaptive Portfolio Governance System <i>Elsevier – Engineering Applications of Artificial Intelligence</i> Focus: agentic AI orchestration, portfolio optimization, explainable governance, market-regime adaptation, instability detection, anti-hallucination safeguards, auditability, and transparent financial decision intelligence.	Under Review, 2026
Agentic AI Portfolio Governance / Financial Intelligence <i>Springer Conference Proceedings</i> Focus: agentic AI, multi-agent portfolio governance, explainable financial intelligence, compliance-aware portfolio decision support, and audit-ready AI workflows.	Under Review, 2026

EDUCATION

M.Tech – Artificial Intelligence and Data Science <i>K J Somaiya College of Engineering Mumbai CGPA: 8.06</i> Thesis: Personal Chatbot for Stock Portfolio Management with Agentic AI Framework. Research focus: agentic AI, multi-agent portfolio governance, explainable AI, quantitative finance, portfolio optimization, market-regime analysis, risk-aware portfolio construction, governance validation, and compliance-aware financial intelligence systems.	Aug 2024 – Apr 2026
B.Tech – Electronics and Communication Engineering <i>Presidency University Bangalore CGPA: 7.77</i> Final-year project: Swarm Robots for Agriculture using embedded systems, image processing, IoT sensors, autonomous coordination, and plant disease detection.	Aug 2019 – Jul 2023
Intermediate, MPC – Sri Chaitanya Junior College CGPA: 9.28	Jun 2017 – Jun 2019
10th Grade – T V N R M National High School CGPA: 9.5	Jun 2016 – Jun 2017

CERTIFICATIONS & TEST SCORES

Certifications: Smart Manager Certificate (CII), Python Programming, Machine Learning, C Programming
Standardized Tests: GATE Computer Science: 23.2 | GATE AI & Data Science: 33.3 | JEE Mains: 85.6 Percentile

LANGUAGES

English: Fluent | Telugu: Native | Hindi: Intermediate